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Indium-free perovskite/silicon tandem solar cells with tin oxide recombination layer and electrodes

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Indium-based transparent conductive oxides are widely used as electrodes and recombination layers in perovskite/silicon tandem solar cells, yet their scalability is constrained by indium scarcity and sputtering-induced damage. Here we report high efficiency and stable indium-free perovskite/silicon tandem solar cells enabled by reactive plasma deposited tin oxide (RPD-SnO_x). For RPD-SnO_x as the recombination layer, a certified efficiency of 33.6% is achieved. Fully indium-free tandems that used RPD-SnO_x as both recombination layer and electrodes delivering a champion PCE of 33.2% (1 cm²) and a mini-module with a certified efficiency of 31.0% (207.9 cm²). Dense and uniform self-assembled monolayer anchoring enabled by RPD-SnO_x suppressed non-radiative recombination and reduced halide migration. Indium-free mini-modules exhibited high thermal, damp-heat, and outdoor operational stability and retained 65% of their maximum initial efficiency after 105 days of outdoor operation.

Monolithic perovskite/silicon tandem solar cells integrate wide-bandgap perovskite top cells with crystalline silicon (*c*-Si) bottom cells (1, 2). High-performance tandem solar cells critically depend on a well-engineered recombination layer and transparent electrodes. These components must provide electrical conductivity to enable efficient electron-hole recombination between adjacent subcells and maintain high transparency to minimize parasitic absorption (3–5). Indium-based transparent conducting oxides (TCOs), particularly indium tin oxide (ITO) and indium zinc oxide (IZO), have been the prevailing choice because of their outstanding optoelectronic properties, including low sheet resistance (*R*_s) and high transmittance (6–12). Jia *et al.* achieved a certified power conversion efficiency (PCE) of 34.6% with an area of 1.002 cm² on tandem using IZO as both the front electrode and recombination layer (13).

Despite these advantages, the reliance on indium poses challenges in terms of cost and sustainability. Indium is a scarce element with limited global reserves and production capacity, resulting in high prices that hinder the commercial scalability of tandems (14). Moreover, magnetron sputtering, the technique commonly used to deposit indium-based TCOs,

can induce damage to underlying functional layers, thereby degrading device performance and yield (15). Together, these limitations highlight the urgent need for indium-free (In-free) alternative films that combine low deposition damage with optoelectronic properties comparable to, or even surpassing, those of conventional TCOs, enhancing both PCE and commercial viability of tandem solar cells.

Prior studies have extensively investigated In-free electrodes for silicon heterojunction (SHJ) solar cells, including a range of zinc- and tin-based TCOs, such as aluminum doped zinc oxide (AZO) (16–20), boron doped zinc oxide (BZO) (21, 22), gallium doped zinc oxide (GZO) (23), tantalum doped tin oxide (TTO) (24), and SnO_x (25–27). In contrast, the progress toward In-free tandems have remained limited because of the scarcity of suitable recombination layers. Doped silicon-tunneling recombination junctions represent one possible approach but suffer from high parasitic absorption loss and require complex processing (28). Alternative interconnection approaches, including TiO_x/TiN_x stack (29) and *n*-doped C₆₀/*p*-doped TaTm junction (30), have also been explored, yet none has so far demonstrated PCEs exceeding established state-of-the-art benchmarks.

Recently, the industrial viability of RPD-SnO_x has been highlighted for its potential application as an electron transport layer in single-junction perovskite solar cells (31). Koida *et al.* further elucidated the pivotal role of regulating oxygen stoichiometry in modulating the optoelectronic properties of RPD-SnO_x (32). Building upon these works, we developed In-free tandem solar cells using RPD-SnO_x with tailored optoelectronic properties as both the recombination layer and transparent electrodes. We present a comprehensive investigation of the optoelectronic properties, interfacial chemistry, charge recombination, and tandem performance enabled by RPD-SnO_x. We achieved a champion PCE of 33.9% (certified 33.6%) for tandems with RPD-SnO_x recombination layer (SnO_x-RL tandem), and 33.2% for In-free tandems with RPD-SnO_x recombination layer and electrodes (In-free tandem) measured with a mask area of 1 cm². An In-free mini-module fabricated from a single industrial-size tandem cell (21 × 10.5 cm²) attained a certified PCE of 31.0% and exhibited exceptional operational and damp-heat stability, retaining 95.2% of its initial PCE after 1,056 hours storage at 85% relative humidity and 85°C.

Performance of tandem solar cells

In monolithic tandem device with a SHJ bottom cell, three indium-based functional layers are typically used: front transparent electrode (F-TCO) under metal grid, recombination layer (RL-TCO) that electrically connects the perovskite top cell with the SHJ bottom cell, and rear electrode (R-TCO) (Fig. 1A). Using double-side submicrometer-textured SHJ bottom cells, we fabricated SnO_x-RL tandems and In-free tandems using RPD-SnO_x recombination layer and electrodes, as well as control device using conventional IZO F-TCO (40 nm), ITO RL-TCO (20 nm) and R-TCO (80 nm). The detailed information for all TCOs is summarized in table S1. High-angle annular dark-field scanning transmission electron microscopy (HAADF-STEM), combined with energy-dispersive x-ray spectroscopy (EDX) elemental mapping, confirmed the thickness of the compact perovskite absorber to be about 800 nm as evidenced by the iodine signal, and resolved the thin functional layers on and under perovskite, including C₆₀, SnO_x, Ag and MgF₂ (Fig. 1B). A conformal SnO_x recombination layer (~15 nm) uniformly coated the submicrometer surface textures, as evidenced by the homogeneous and continuous distribution of Sn and O elements (fig. S1). The RPD-SnO_x front electrode (50 nm) and the ALD-SnO_x (15 nm) buffer layer formed an intimately integrated bilayer (~65 nm) without a discernible interface, indicating excellent interlayer compatibility. Cross-sectional scanning electron microscopy (SEM) further reveals high-quality perovskite compactly grown on the textured SHJ bottom cell, characterized by vertically aligned, large grains spanning the full film thickness (fig. S2).

The illuminated current density-voltage (*J-V*) characteristics of champion devices revealed that the SnO_x-RL tandem achieved a PCE of 33.9% (1 cm²), whereas the In-free tandem attained a slightly lower PCE of 33.2%. Both values exceeded that of the control device (32.7%) (Fig. 1C). Stabilized maximum power point (MPP) tracking further confirmed PCEs of 33.8%, 33.1% and 32.6%, respectively, under standard AM1.5G spectrum for 300 s. The independent certification yielded a PCE of 33.6% for the champion SnO_x-RL tandem, with an open-circuit voltage (*V*_{oc}) of 1.98 V, a short-circuit current density (*J*_{sc}) of 20.8 mA/cm² and a fill factor (FF) of 81.7% (Fig. 1D and fig. S3). The external quantum efficiency (EQE) spectrum of the SnO_x-RL tandem gave integrated current densities of 21.7 and 20.7 mA/cm² for the perovskite top cell and the SHJ bottom cell, respectively, in good agreement with the measured *J*_{sc} (Fig. 1E). Device-to-device statistics collected from 20 cells of each configuration are summarized in the PCE box plots and the corresponding photovoltaic parameters (Fig. 1F and fig. S4). The SnO_x-RL tandems exhibited an average PCE of 33.3% (1 cm²), substantially higher than that of the control devices (32.2%). By comparison, the In-free tandems showed a slightly lower average PCE of 32.7%.

The enhanced *V*_{oc} of the SnO_x-RL tandem originated from the reduced deposition-induced damage to the SHJ bottom cell and suppressed nonradiative recombination at the buried interface, enabled by the stronger bidentate anchoring, higher coverage density and more uniform dipole formation of self-assembled monolayer (SAM) on SnO_x-RL. The higher FF we attributed primarily to more efficient and homogeneous charge recombination at the perovskite/SAM/SnO_x interface, enabled by favorable band alignment at the SnO_x/nc-SiO_x:H(n) interface with smaller barrier and reduced interfacial recombination losses. The slightly higher *J*_{sc} of the SnO_x-RL tandem is attributed to the reduced parasitic absorption and reflection losses at the long wavelength region, as confirmed by the optical loss analysis using a ray-tracing simulations (fig. S5) (33). When SnO_x is further employed to replace the front IZO and rear ITO in In-free device, parasitic absorption increases, leading to a reduction in current for both the perovskite top cell and the SHJ bottom cell, respectively. However, the SHJ bottom cell current, the limiting current in the tandem, remain slightly higher than that of the SHJ sub-cell in the control device. This implies that although we have increased the thickness of the front SnO_x to 50 nm to minimize its *R*_s (132 ohm/sq) in In-free devices, the parasitic absorption penalty remained unavoidable due to its lower transmittance compared to IZO (fig. S6). Despite the optimization, the *R*_s of the front SnO_x is still slightly higher than that of IZO (118 ohm/sq). This, coupled with the elevated contact resistivity at the rear SnO_x/nc-Si:H(p) (34), constitutes the primary origin for a lower FF in In-free devices (fig. S7).

The operational stability of the unencapsulated tandem solar cells was evaluated by maximum power point (MPP) tracking under continuous 1-sun illumination in nitrogen atmosphere at room temperature (Fig. 1G). The control device degraded to $\sim 50\%$ of its initial PCE after 3,000 hours operation. In contrast, the SnO_x -RL tandem retained 70.8% of its initial PCE, whereas the In-free tandem exhibited enhanced stability, maintaining 81.1% of its initial PCE. The improved operational stability of the In-free tandem compared to the SnO_x -RL tandem and the control device arose from the combined effects of robust interfacial chemistry and effective suppression of ion migration. SnO_x -based devices benefitted from stronger SAM binding and reduced interfacial defect density enabled by the SnO_x -RL, and the In-free tandem further eliminates indium-based TCOs that contain nanocrystalline domains prone to halide diffusion (35). Fundamentally, in conventional TCOs, grain boundaries and defective amorphous-crystalline interfaces act as fast channels for ion migration. In contrast, the amorphous RPD- SnO_x electrodes eliminates these pathways, thereby effectively block iodide migration from the perovskite film to the surface and mitigate Ag electrode corrosion. Thus, the In-free tandem retained the highest fraction of its initial PCE under prolonged operation, followed by the SnO_x -RL tandem, whereas the ITO/IZO-based control device degraded most rapidly.

RPD- SnO_x recombination layer

The recombination layer electrically connects the perovskite top cell with the *c*-Si bottom cell (4) and enables efficient recombination of photogenerated holes and electrons across a SAM-modified interface (36). Our previous work has demonstrated RPD- SnO_x as an effective transparent electrode for SHJ solar cells, with tunable optoelectronic properties and reduced deposition damage to underlying functional layers (24, 34). We adapted this approach to engineer SnO_x recombination layers for tandems with high carrier concentration by reducing the oxygen flow ratio during the RPD process. Hall effect measurements showed that the RPD- SnO_x recombination layer shown higher carrier concentration ($1.18 \times 10^{21} \text{ cm}^{-3}$) and lower carrier mobility ($1.4 \text{ cm}^2/\text{Vs}$) than sputtered ITO (table S2). SHJ bottom cells incorporating the RPD- SnO_x recombination layer displayed a higher implied V_{oc} than that with sputtered ITO (fig. S8), which partly accounted for the enhanced V_{oc} observed in the tandems. We subsequently investigated the impact of SnO_x -RL thickness on tandem performance. The results revealed that the devices with a 15 nm SnO_x -RL achieve the maximum PCE, which is attributed to the superior V_{oc} and FF (fig. S9). This represents a delicate balance between the optoelectronic properties of RPD- SnO_x recombination layer and its synergistic impact on the overall performance of the tandem solar cells (figs. S10 and S11).

The interfacial charge transfer, device performance, and operational stability of tandem solar cells are strongly influenced by the coverage density and uniformity of SAMs on the recombination layer (9). We studied interfacial interaction by performing density functional theory (DFT) calculations that investigated the adsorption characteristics of the SAM molecule, [4-(3,6-dimethoxy-9H-carbazol-9-yl)ethyl] phosphonic acid (MeO-4PACz), on SnO_x and ITO surfaces. The calculated charge density difference maps revealed more delocalized and spatially homogeneous charge redistribution for MeO-4PACz adsorbed on SnO_x than on ITO (Fig. 2A), where the cyan and yellow isosurfaces denote electron depletion and accumulation, respectively. Notably, the phosphonic acid anchoring group of MeO-4PACz exhibited bidentate coordination with exposed Sn atoms on the SnO_x surface, whereas predominantly monodentate binding was observed on the ITO surface (fig. S12). Consistent with this trend, the adsorption energy of MeO-4PACz on SnO_x ($E_{\text{ad}} = -6.779 \text{ eV}$) was much larger than that on ITO ($E_{\text{ad}} = -2.323 \text{ eV}$), indicating stronger interfacial binding on SnO_x . To further quantify the bonding strength, crystal orbital Hamilton population (COHP) analysis was conducted (Fig. 2B). The integrated COHP (ICOHP) values for the two O-Sn bonds were -2.74 and -3.18 eV , respectively, and comparable to the ICOHP value of -3.25 eV for the O-In interaction. These results indicated strong orbital interactions between the phosphonate anchoring groups and the SnO_x surface, supporting the formation of robust and stable chemical bonding between SnO_x and MeO-4PACz.

To compare the adsorption capacity of MeO-4PACz on SnO_x and ITO, we used x-ray photoelectron spectroscopy (XPS) to examine the surface stoichiometry after SAM anchoring (fig. S13). The C 1s core-level spectra of SAM-modified SnO_x and ITO each exhibited four characteristic components, among which the C-N and C-P signals provide direct evidence for anchoring of MeO-4PACz on both substrates. Deconvolution of the O 1s spectra revealed four contributions corresponding to lattice metal-oxygen bonds (Sn-O or In-O), oxygen-phosphorus bonds (O-P), hydroxyl groups (-OH), and carbon-oxygen species (C-O). Notably, the O-P/Sn-O intensity ratio for the SnO_x /SAM interface (0.78) was higher than the corresponding O-P/In-O ratio for the ITO/SAM interface (0.69) (table S3), indicating a higher SAM coverage density and stronger interfacial bonding mediated by the phosphonic acid anchoring groups on the SnO_x surface (37, 38). The reason lies in that the exposed surface of the bare SnO_x possesses a higher proportion of -OH than ITO (fig. S14), which act as effective binding sites, facilitating SAM anchoring on the SnO_x substrate. Collectively, these results indicated that the higher density of chemically active anchoring sites on SnO_x enables more uniform SAM coverage, leading to improved interfacial potential homogeneity.

Kelvin probe force microscopy (KPFM) was used to assess the **surface potential distribution** of SnO_x and ITO before and after MeO-4PACz anchoring. The pristine SnO_x surface exhibited a higher local surface potential than ITO (fig. S15). After SAM modification, the SnO_x /SAM interface shows a lower average surface potential compared with the ITO/SAM interface, indicating a pronounced surface dipole induced by the anchored molecules. These trends were consistent with the corresponding work function (WF) values determined from ultraviolet photoelectron spectroscopy (UPS) (Fig. 2C). The WF of SnO_x increased from 4.68 eV to 5.23 eV upon MeO-4PACz anchoring, whereas the WF of ITO rose from 4.64 eV to 5.05 eV. **The higher WF at the SAM/ SnO_x interface favored carrier extraction and recombination, and helps improved tandem performance.** The MeO-4PACz-modified SnO_x exhibited a larger WF shift (0.55 eV) than the corresponding ITO surface (0.41 eV), indicative of a higher interfacial dipole density at the SnO_x /MeO-4PACz interface (9). This observation further supported the presence of a higher density of chemically active anchoring sites on SnO_x , which promoted more ordered SAM orientation and packing (39). Consistent with this interpretation, the SnO_x /SAM interface displayed a narrower contact potential difference distribution than the ITO/SAM interface (fig. S16) that reflected a more uniform SAM coverage (40).

We further simulated band diagrams of the tandems using SCAPS-1D under dark thermal equilibrium conditions, with the key input parameters summarized in tables S4 to S6. These simulations provided qualitative insights into the interfacial electrostatics and equilibrium carrier transport barriers. At the recombination junction, efficient bidirectional carrier extraction is essential for achieving high-performance tandem solar cells (41). The dark-equilibrium band diagrams revealed a lower interfacial energy barrier at the SnO_x /nc-SiO_x:H(n) junction compared with the ITO/nc-SiO_x:H(n) counterpart (Fig. 2, D and E), indicating more favorable electron transport across the SnO_x -based interface. The lower electron affinity of SnO_x (4.4 eV) relative to ITO (4.9 eV) led to improved conduction-band alignment with nc-SiO_x:H(n) (fig. S17). When SnO_x was used as the rear electrode (R-TCO), its lower electron affinity led to less favorable band alignment with *p*-type hydrogenated nanocrystalline silicon (nc-Si:H(p)) for hole extraction from the SHJ bottom cell (fig. S18). This interfacial limitation partly contributes to the inferior photovoltaic parameters (especially FF) observed in the In-free tandems compared with the SnO_x -RL tandems.

Perovskite grown on RPD- SnO_x /MeO-4PACz stack

We systematically investigated perovskite films grown on SnO_x /MeO-4PACz and ITO/MeO-4PACz stacks. The steady-state photoluminescence (PL) intensity of perovskite films deposited on SnO_x /MeO-4PACz was higher than that on

ITO/MeO-4PACz (Fig. 3A), indicating a reduced density of non-radiative recombination centers. Transient photoluminescence (TrPL) probed the charge-carrier dynamics of perovskite films on the different substrates (Fig. 3B). The differential lifetime, $\tau = -\{d \ln[\Phi(t)]/dt\}^{-1}$, where $\Phi(t)$ denotes the time-dependent PL photon flux, was derived from tri-exponential fits to the **TrPL data** (42, 43). The perovskite grown on SnO_x /MeO-4PACz exhibited faster initial charge extraction and a prolonged asymptotic lifetime exceeding 5 μs , compared with a lower lifetime of 3 μs for the perovskite on ITO/MeO-4PACz (Fig. 3C). This pronounced extension of the TrPL lifetime indicated that non-radiative recombination losses at the buried interface were suppressed.

To quantitatively assess non-radiative recombination losses at the buried interface, **photoluminescence quantum yield (PLQY) measurements were performed** (44). The PLQY of the perovskite grown on SnO_x /MeO-4PACz reached 0.223%, corresponding to a quasi-Fermi level splitting (QFLS) of 1.255 eV, whereas the corresponding values for the film on ITO/MeO-4PACz are 0.116% and 1.238 eV, respectively (Fig. 3, D and E). The higher QFLS indicates improved interfacial energy-level alignment at the SnO_x /SAM/perovskite interface, which reduced the defect density and promoted more efficient carrier recombination. These optimized carrier dynamics ultimately translate into enhanced V_{oc} and FF in the tandem solar cells. **Intensity-dependent QFLS measurements further revealed a lower ideality factor of 1.45 for perovskite grown on SnO_x /MeO-4PACz, compared with 1.53 for those on the ITO/MeO-4PACz stack** (Fig. 3F). This reduction in ideality factor pointed to suppressed Shockley-Read-Hall (SRH) recombination that resulted from a diminished population of deep-level trap states at the buried interface (45, 46). We further reconstructed the **pseudo-*J-V*** for control and SnO_x -RL tandem solar cells, which is carried out under open-circuit conditions and free of transport losses (fig. S19). This analysis directly revealed the origins of the performance difference. The perovskite top cell on the SnO_x exhibits a higher pV_{oc} , which confirms its superior interface quality and lower non-radiative recombination loss. Meanwhile, the difference between its pFF and FF is 4.09%, which is lower than the 4.89% observed for the control device, indicating a lower series-resistance loss. These results confirm that SnO_x -RL synergistically enhanced both the V_{oc} and FF by suppressing non-radiative recombination and reducing electron transport loss.

High-quality perovskite films are indispensable for efficient tandem solar cells, but stabilizing their elemental composition is equally critical for sustaining long-term operational performance. We used time-of-flight secondary ion mass spectrometry (TOF-SIMS) to compare elemental distributions in fresh and aged devices, with aging performed for 1,000 hours under continuous 1-sun illumination in N_2 atmosphere at room temperature (fig. S20). The elemental

depth profiles revealed that, in the aged control device, a substantial fraction of iodide ions penetrated through the IZO front electrode and accumulated within the MgF_x antireflection coating (Fig. 3G). In contrast, the iodide distribution in the aged In-free tandem remained nearly identical to that of its fresh counterpart (Fig. 3H). We attributed the enhanced stability of the In-free tandems to the high damp-heat stability of the amorphous SnO_x layers (26, 34), which effectively reduced ion-migration pathways and suppress halide diffusion and interfacial corrosion. By comparison, the widely-used IZO front electrode consisted of numerous nanocrystallites embedded within an amorphous matrix (fig. S21) and was less effective in blocking halide-ion migration (47). Because perovskite decomposition is partially driven by the moisture-assisted generation of mobile halogen (48, 49), the synergistic barrier formed by the ALD- SnO_x and front electrode effectively mitigated this degradation pathway. Photographs of the aged devices clearly illustrate that, compared to the In-free counterpart, the silver electrodes of the control device appear dull and yellowish (fig. S22). A signature peak of silver iodide (AgI) emerged at 23.7° in the XRD patterns of aged control device (fig. S23), indicating that outward-migrating iodide ions penetrated the front IZO and reacted with the silver electrode. In the In-free device, RPD- SnO_x not only serve as a barrier that protects the perovskite and mitigates its decomposition, but also effectively obstructs the migration pathways of iodide ions, thereby shielding the Ag electrode from chemical erosion, resulting in improved device stability.

Industrial-size In-free tandem solar cells

We further validate the In-free tandem solar cells at an industrial scale using slot-die-coated perovskite films and an otherwise similar device architecture ($21 \times 10.5 \text{ cm}^2$) (fig. S24). The RPD- SnO_x electrodes deposited under a higher oxygen flow ratio shown reduced carrier concentration but higher carrier mobility (table S2). The champion In-free tandem mini-module (active area of 207.9 cm^2), fabricated from a single industrial-size device, achieved a certified PCE of 31.0% (Fig. 4A and fig. S25). The device yielded a stabilized power output of 6.39 W under MPP tracking, confirming a high PCE of 30.7% (fig. S26). Compared with laboratory-scale tandems, the PCE of the mini-module was primarily limited by a dramatically reduced J_{sc} of 18.9 mA/cm^2 . The current densities integrated from the EQE spectra are 20.3 and 19.1 mA/cm^2 for the perovskite top cell and the SHJ bottom cell, respectively (Fig. 4B). Due to the localized beam spot size of the EQE measurement, the shading and reflection losses associated with the screen-printed fingers and ribbons are not captured in the integrated current density. This discrepancy indicated that in addition to parasitic absorption in the SnO_x transparent electrodes and encapsulation layers, reflection and shading losses associated with the encapsulated glass and screen-

printed metal fingers constituted major contributors to the reduced current density in both sub-cells. To gain a deeper understanding of the cell-to-module current losses, we performed optical simulations on both the In-free small-area cell and the mini-module by combining ray tracing and transfer matrix methods (fig. S27). The results reveal that during the upscaling process from cell to mini-module, the primary reasons for the current reduction are the increased reflection loss caused by the module encapsulation (glass and TPO), and the front optical shading caused by the interconnection ribbons and silver grids. Conversely, the parasitic absorption contributed by the SnO_x recombination layer is substantially minimized and does not constitute a limiting factor for modules. The V_{oc} of 1.94 V was lower than that of laboratory-scale tandems (1.97 V), likely associated with the larger spatial variation in defect density and interfacial quality that increased non-radiative recombination and reduced average QFLS. However, the FF increased to 84.3%, which partially compensated for the losses in J_{sc} and V_{oc} . This may arise from the pronounced current mismatch that shifts the operating point toward a higher FF regime (50), together with a combination of reduced shunt losses, edge recombination and optimized grid design enabled efficient lateral charge transport.

To evaluate the long-term stability of the In-free tandem mini-modules, a series of complementary stability tests were conducted following the ISOS protocols. First, the storage stability of mini-module was examined in N_2 box following the ISOS-D-1 protocol. The unencapsulated devices were stored in the dark under a controlled N_2 atmosphere and a relative humidity (RH) of below 0.001% at room temperature, with their PCEs measured at 12-day intervals. Both the In-free and SnO_x -RL tandems exhibit excellent storage stability, retaining 99.0% and 91.4% of their initial PCEs after 3,168 hours of storage, respectively (Fig. 4C). The control device retains only 79.5% of its initial PCE over the same period. Subsequently, MPP tracking of the In-free and control mini-modules was performed at 85°C under continuous 1-sun illumination in N_2 atmosphere (ISOS-L-3, Fig. 4D). After 1,000 hours, the In-free device retained 93.9% of its initial PCE, demonstrating high operational stability under elevated-temperature conditions. In addition, the In-free tandem mini-module shown slight PCE degradation (-4.8%) after 1,056 hours test under damp-heat conditions following the ISOS-D-3 protocol (85% relative humidity and 85°C) (Fig. 4E), confirming its exceptional resistance to combined thermal and moisture stress.

Finally, encapsulated In-free and control devices were deployed for outdoor field testing. After approximately 105 days of operation, including exposure to multiple rainy and overcast days, the In-free mini-module retained 65% of its maximum initial PCE, whereas the control device underwent rapid degradation to 5% of its maximum initial PCE after approximately 75 days (Fig. 4F). The improved stability of the

In-free tandem mini-modules stemmed from the synergistic effects of robust interfacial chemistry, effective ion-migration blocking and intrinsically stable SnO_x electrodes, mitigating non-radiative recombination, interfacial degradation and halide ion migration toward the electrodes during long-term operation. In addition, the reduced current density and improved thermal dissipation in the mini-module further alleviate electrical and thermal stress. Thus, performance trade-offs associated with upscaling are fundamentally manageable and do not preclude high-efficiency tandem operation.

Conclusions

We demonstrated a viable and scalable route toward high-efficiency and stable In-free tandem solar cells by replacing indium-based TCOs with RPD- SnO_x . Beyond achieving state-of-the-art PCEs at both small and large scales, this work revealed that RPD- SnO_x enables superior interfacial quality through dense SAM anchoring, suppressed non-radiative recombination, and mitigated halide migration. The tandems not only exhibited high PCEs but also exceptional operational, thermal and outdoor stability. By decoupling high performance from reliance on scarce and costly indium, this study establishes RPD- SnO_x as a broadly applicable electrode and recombination layer for scalable, high-efficiency and durable tandem photovoltaics.

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SUPPLEMENTARY MATERIALS

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Materials and Methods

Figs. S1 to S29

Tables S1 to S9

References (51–64)

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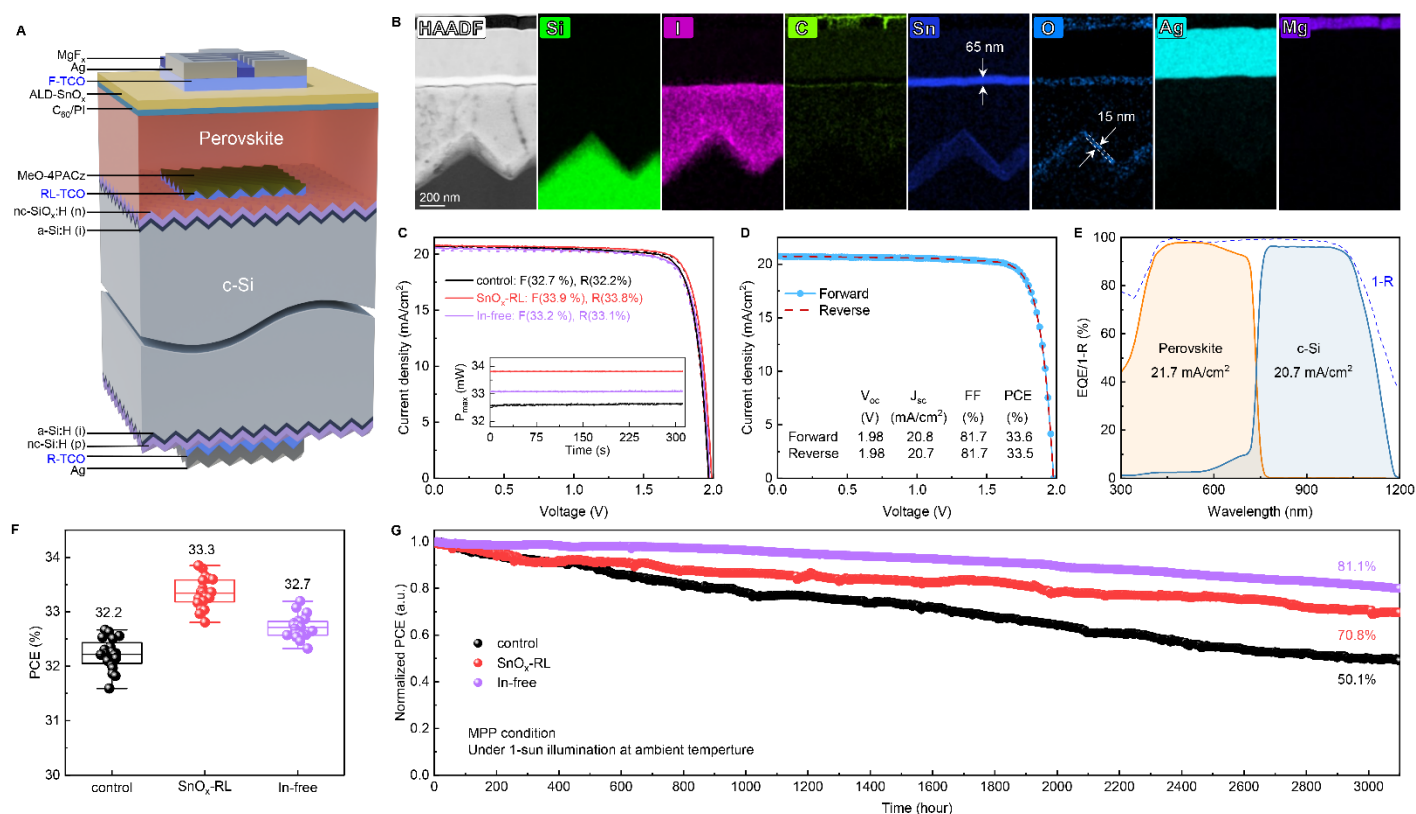


Fig. 1. Tandem performance. (A) Schematic of monolithic perovskite/c-Si tandem solar cell. (B) HAADF-STEM images with EDX mapping of the In-free tandem. (C) Illuminated $J-V$ curves of the control, SnO_x-RL and In-free tandems under AM1.5G illumination. The inset shows the stabilized power output for 300 s. (D) The certified $J-V$ and (E) EQE curves of the SnO_x-RL tandem (1.002 cm²). (F) The PCE box plot of the control, SnO_x-RL and In-free tandems. Twenty devices were fabricated for each type. (G) MPP tracking of the unencapsulated tandem cells under continuous 1-sun illumination in N₂ atmosphere.

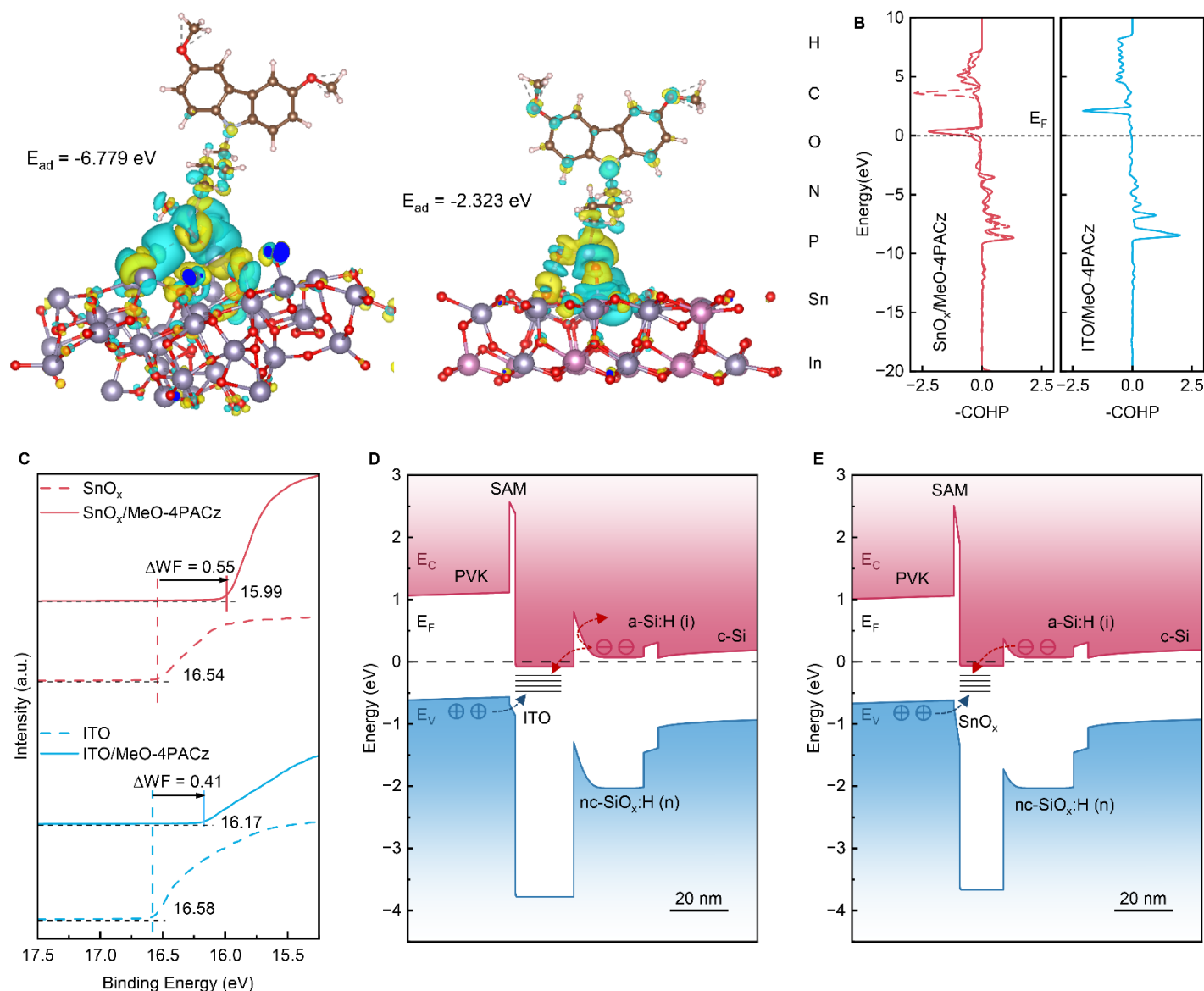


Fig. 2. Adsorption of MeO-4PACz on RPD-SnO_x. (A) DFT calculation of MeO-4PACz adsorbed on SnO_x and ITO, and charge transfer between MeO-4PACz and SnO_x (ITO). (B) COHP curves of Sn–O and In–O bonds. (C) UPS spectra of ITO and SnO_x with and without MeO-4PACz anchoring. (D and E) Simulated band diagrams of the tandems with (D) ITO and (E) RPD-SnO_x recombination layers.

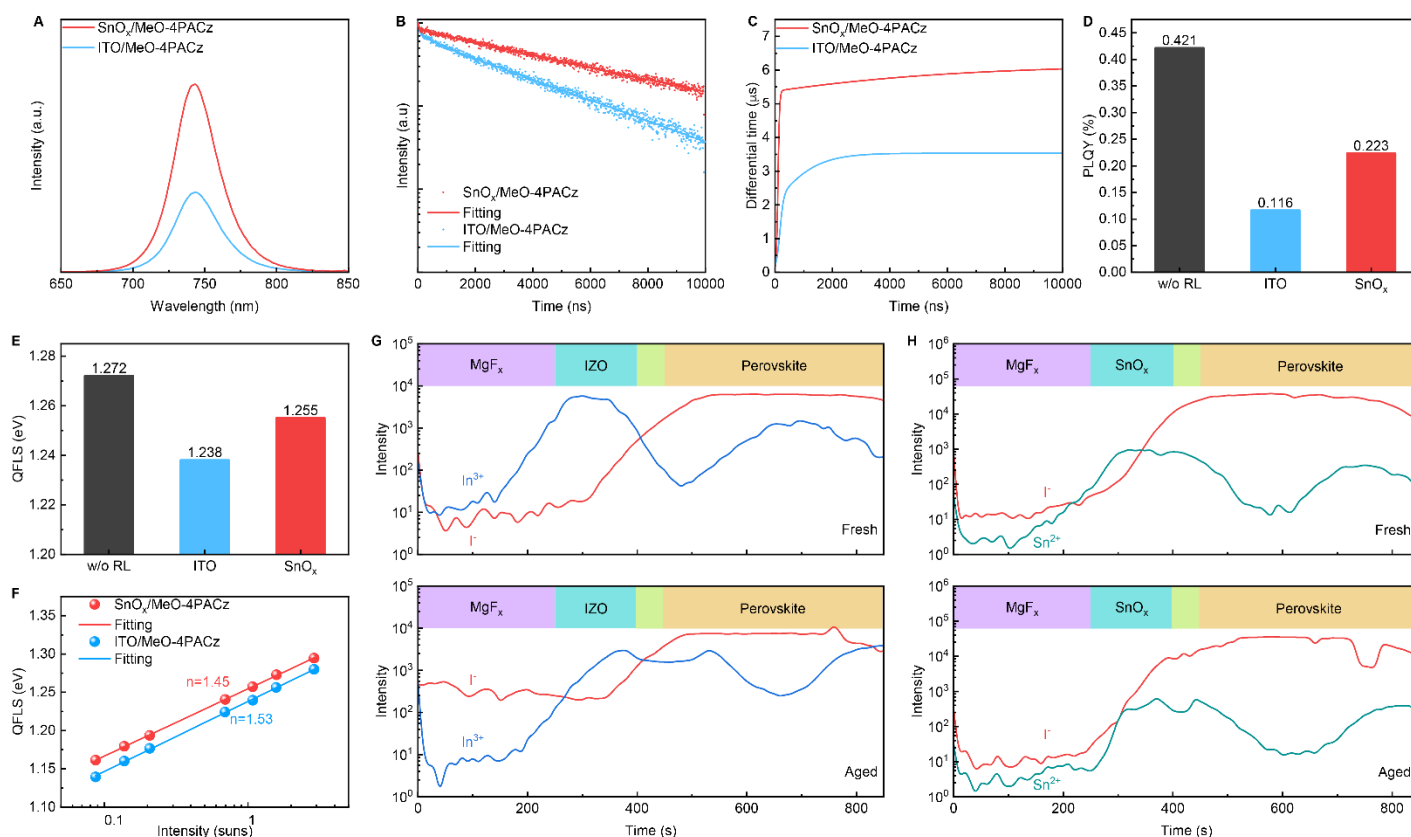


Fig. 3. Perovskite grown on RPD-SnO_x/MeO-4PACz. (A) Steady-state PL spectra, (B) transient PL spectra and (C) corresponding fitted differential lifetimes of perovskites grown on ITO/MeO-4PACz and SnO_x/MeO-4PACz. (D) PLQY and (E) QFLS of perovskites grown on MeO-4PACz coated ITO and SnO_x recombination layers. (F) The corresponding V_{oc} evolution as a function of light intensities. (G and H) TOF-SIMS depth profiles of I⁻ in fresh and aged (G) control and (H) In-free tandems.

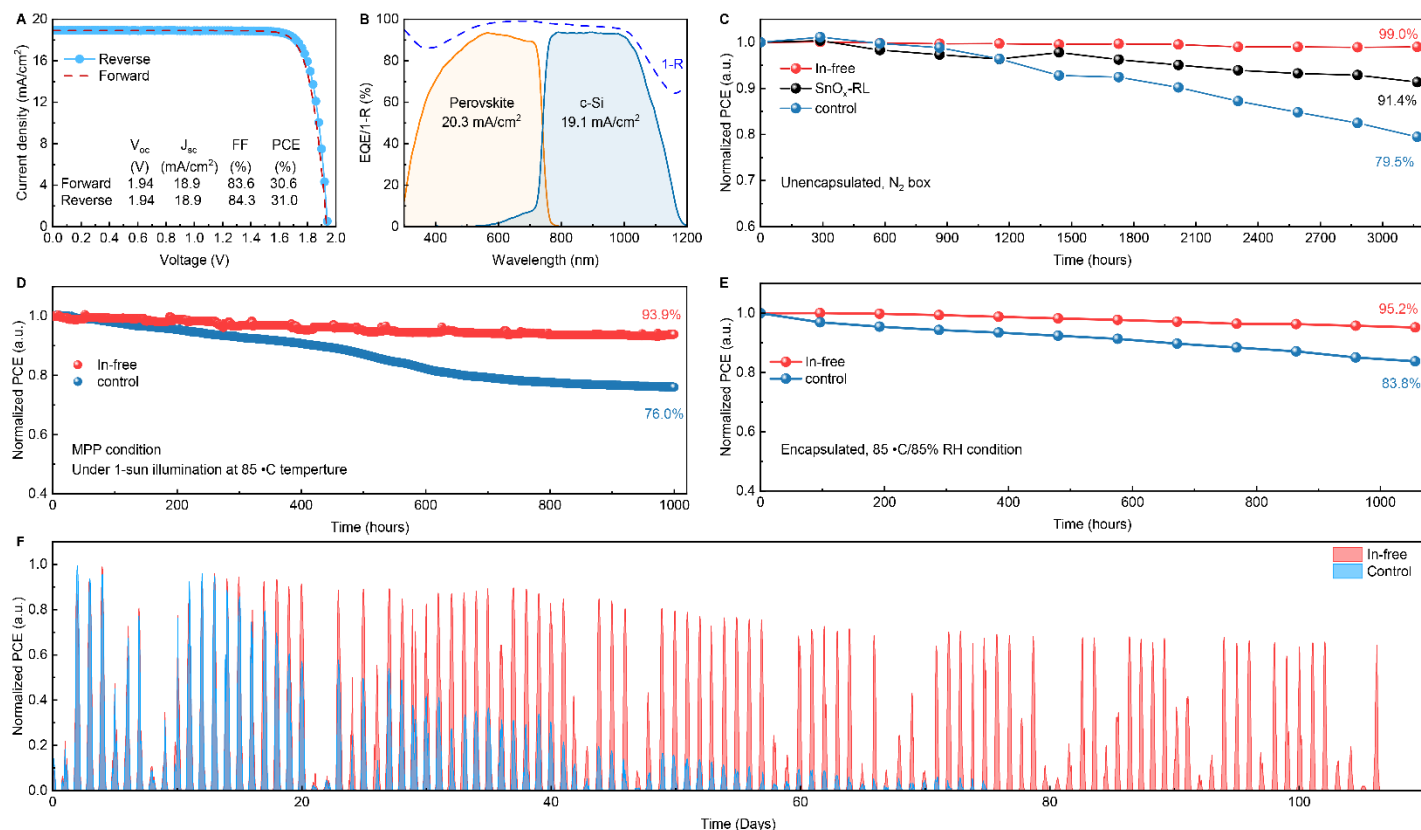


Fig. 4. Performance of industrial-size In-free tandems. (A) Certified J - V curves and (B) EQE of the In-free tandem mini-module (designated area 207.9 cm²). (C) Storage stability of the unencapsulated In-free, SnO_x-RL and control tandems in N₂ atmosphere. The device parameters of mini-modules were list in table S7. (D) MPP tracking of the encapsulated control and In-free mini-modules at 85°C under continuous 1-sun illumination. The device parameters of mini-modules were list in table S8. (E) Damp-heat test of control and In-free mini-modules at 85°C/85% RH. The device parameters of mini-modules were list in table S9. (F) Outdoor performance of the In-free and control tandem mini-modules measured at Soochow University in Jiangsu Province, China for 105 days.



Indium-free perovskite/silicon tandem solar cells with tin oxide recombination layer and electrodes

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